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Department II



Research Poster Appendix

Supporting Research Documentation

Same Meaning, Different Answer?

Evaluating Answer Consistency in Large Language Models Across
Linguistic Transformations

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Contents of the Supporting Appendix

1	Introduction and Contribution	1
1.1	Background and Research Motivation	1
1.2	Research Questions	1
1.3	Research Contribution	2
2	Related Work	2
2.1	Linguistic Variation and Model Robustness	2
2.2	Presentation Sensitivity and Research Gap	2
3	Experimental Methodology	2
3.1	Datasets and Experimental Sample	2
3.2	Linguistic Transformations	3
3.3	Example of a Question Group	3
3.4	Transformation Validation	3
4	Models and Evaluation Methodology	4
4.1	Models and Answer Extraction	4
4.2	Evaluation Metrics	4
4.3	Statistical Analysis	5
5	Results and Analysis	5
5.1	Accuracy Across Formulations	5
5.2	Original and Strict Robust Accuracy	6
5.3	Correctness Flips and a Real Example	6
5.4	Statistical Findings	8
5.5	Findings by Research Question	8
6	Discussion and Limitations	8
6.1	Interpretation of the Findings	8
6.2	Limitations and Future Work	9
7	Conclusion and Reproducibility	9
7.1	Conclusion	9
7.2	Reproducibility and Research Materials	9
	References	10
	Declaration of Authenticity	12

List of Figures

1	Accuracy by linguistic formulation	6
2	Original versus strict robust accuracy	7
3	Correctness flip rates	7

List of Tables

1	Evaluation dataset composition	3
2	Example of linguistic transformations	3
3	Evaluated models and checkpoints	4
4	Accuracy by linguistic formulation	5
5	Original versus strict robust accuracy	6
6	Correctness transitions and flip rates	7
7	Example of a correctness change	8

Abstract

Large language models (LLMs) are usually evaluated on questions written in one fixed form. However, a model may answer a question correctly and give a wrong answer when the same question is written differently. This study examines whether LLMs maintain correct answers when the wording, sentence structure, or order of information changes.

The experiment uses 30 original questions from CommonsenseQA, GSM8K, and LogiQA. Each question is tested in its original form and in three rewritten versions: Lexical, Syntactic, and Information Order. The rewritten questions are intended to preserve the same meaning and correct answer. All 90 rewrites passed automated structural checks, and 45 were also manually reviewed. Three open-weight models were evaluated: Qwen2.5-3B-Instruct, Llama-3.2-3B-Instruct, and Gemma-3-1B-IT, producing 360 responses.

Performance is measured using accuracy, correctness changes, flip rates, and strict robust accuracy. Strict robust accuracy requires a model to answer all four versions of a question correctly. It was 30.0% for Qwen, 26.7% for Llama, and 20.0% for Gemma, compared with original-question accuracies of 63.3%, 50.0%, and 23.3%, respectively. No tested comparison remained statistically significant after multiple-testing correction.

The findings show why evaluating only the original question may give an incomplete picture of model performance.

Keywords: Large Language Models, Linguistic Robustness, Meaning-Preserving Transformations, Answer Correctness, Correctness Flips, Strict Robust Accuracy

GitHub: <https://github.com/Yash-Gavade/ReasonShift>

1 Introduction and Contribution

1.1 Background and Research Motivation

Large language models (LLMs) are often evaluated using questions that test commonsense, mathematical, or logical reasoning. Usually, each question is presented in one fixed form. However, a model may answer the original question correctly and give a wrong answer when the same question is rewritten. Earlier research has shown that changes in wording can reveal differences in model behavior [Ribeiro et al., 2018; Srikanth et al., 2024].

This matters because overall accuracy does not show whether a model answers the same individual questions correctly across different formulations. For example, a model may lose one correct answer but gain another, leaving its overall accuracy unchanged. Testing alternative formulations helps identify these changes [Ribeiro et al., 2020].

This study compares each original question with three versions intended to preserve the same meaning and correct answer: Lexical, Syntactic, and Information Order. The aim is to examine whether final-answer correctness remains stable when the language changes.

1.2 Research Questions

The study addresses three research questions:

1. **RQ1 – Performance Change:** How does accuracy change under lexical, syntactic, and information-order reformulations?
2. **RQ2 – Correctness Stability:** How often does rephrasing change whether a model’s answer is correct or incorrect?
3. **RQ3 – Robust Accuracy:** How often does a model answer all four versions of the same question correctly, compared with its original-question accuracy?

The study tests whether changes in linguistic form can change answer correctness even when the intended problem and reference answer remain the same.

1.3 Research Contribution

The study makes three contributions. First, it creates a paired evaluation set containing original reasoning questions and three rewritten versions of each question. Second, it evaluates three open-weight language models using the same inputs and generation settings.

Third, it combines accuracy with correctness flips and strict robust accuracy. These measures show not only how many answers are correct, but also whether correctness is maintained across different versions of the same question.

The study evaluates final-answer correctness. It does not investigate the models’ internal reasoning or the faithfulness of their generated explanations.

2 Related Work

2.1 Linguistic Variation and Model Robustness

Previous research has shown that NLP models may respond differently when the same information is expressed in different ways. Ribeiro et al. studied semantically equivalent changes and introduced CheckList to test model behavior beyond overall accuracy [Ribeiro et al., 2018; Ribeiro et al., 2020].

More directly related to this study, Srikanth et al. examined how paraphrasing relates to errors in natural-language reasoning [Srikanth et al., 2024]. Other work has investigated different paraphrase types and robustness under non-adversarial input changes [Wahle et al., 2024; Ackerman et al., 2024].

These studies show why a correct answer to one version of a question may not be enough to understand a model’s behavior.

2.2 Presentation Sensitivity and Research Gap

The way information is presented can also affect model answers. Previous studies have examined the order of multiple-choice options and the position of relevant information in long texts [Pezeshkpour and Hruschka, 2024; Liu et al., 2024]. These settings differ from the present study, which rearranges existing facts within a question while aiming to preserve its meaning and correct answer.

Research on prompt sensitivity further shows why answer evaluation must be handled carefully [Hua et al., 2025].

Building on this work, the present study compares each original reasoning question with three rewritten versions: Lexical, Syntactic, and Information Order. It measures not only accuracy but also changes from correct to incorrect answers, changes in the opposite direction, and correctness across all four versions.

This paired approach helps identify changes in individual answers that a single accuracy score may hide.

3 Experimental Methodology

The experiment compares each original question with three rewritten versions intended to preserve the same meaning and correct answer. This allows us to check whether a model’s correctness changes when the question is expressed differently [Ribeiro et al., 2018; Srikanth et al., 2024].

3.1 Datasets and Experimental Sample

The study uses ten original questions from each of three benchmarks: CommonsenseQA for commonsense reasoning [Talmor et al., 2019], GSM8K for mathematical word problems [Cobbe et al., 2021], and LogiQA for logical reasoning [Liu et al., 2020].

For each source question, we generated three meaning-preserving reformulations while retaining the

original question as a reference. The four versions share a question-group identifier and the same intended correct answer. The final dataset therefore contains 30 groups and 120 inputs, as shown in Table 1.

Table 1: Composition of the evaluation dataset. Each original question has three rewrites.

Dataset	Original	Rewritten	Total
CommonsenseQA	10	30	40
GSM8K	10	30	40
LogiQA	10	30	40
Total	30	90	120

3.2 Linguistic Transformations

Three types of rewriting were used [Wahle et al., 2024]:

- **Lexical:** Replace words or phrases with suitable alternatives while keeping the meaning.
- **Syntactic:** Change the grammatical structure or order of clauses.
- **Information Order:** Rearrange existing facts or conditions without intending to add or remove information.

Information Order refers to facts within the question, not the order of multiple-choice answer options. The latter is a different type of variation studied in earlier work [Pezeshkpour and Hruschka, 2024].

3.3 Example of a Question Group

Table 2 shows an example question group derived from the CommonsenseQA benchmark. The example illustrates the applied transformation process in the final evaluation dataset (**RS_COMMONSENSE_QA_001**).

Table 2: Example of a CommonsenseQA benchmark item and its three rewritten versions used in the evaluation dataset.

Version	Question
Original	Where can you put a picture frame when it’s not hung vertically?
Lexical	Where can you place a picture frame when it is not hanging vertically?
Syntactic	When a picture frame is not hung vertically, where can it be placed?
Information Order	A picture frame is not being hung vertically. Where can you put it?

All four variants retain the same answer choices: A. art show; B. wall; C. newspaper; D. car; E. table. The intended correct answer is **E. table**.

The lexical transformation replaces “put” with “place”. The syntactic transformation changes the position of the condition, while the information order transformation presents the condition before the question.

The example demonstrates how the same reasoning task is expressed through different linguistic forms while preserving the intended answer.

3.4 Transformation Validation

A rewrite must preserve the intended question and correct answer. Otherwise, a change in model correctness might be caused by a different problem rather than different wording.

The final 90 rewrites passed automated structural checks for text change, answer-choice preservation where applicable, and numerical preservation where applicable.

A stratified sample of 45 rewrites was also reviewed manually for meaning and answer preservation, transformation type, fluency, and answer leakage. All 45 reviewed rewrites were accepted.

The other 45 rewrites passed the automated checks but did not receive the same documented manual semantic review. Automated acceptance alone does not prove that every rewrite preserves meaning.

The final dataset, transformation instructions, and validation records are available in the GitHub repository under `data/validated/`, `prompts/`, `annotations/`, and `audit/`.

4 Models and Evaluation Methodology

4.1 Models and Answer Extraction

Three open-weight, instruction-tuned models were evaluated using the same 120 inputs: Qwen2.5-3B-Instruct [Qwen, n.d.], Llama-3.2-3B-Instruct [Meta, n.d.], and Gemma-3-1B-IT [Google, n.d.]. Their exact checkpoints are shown in Table 3.

Table 3: The three evaluated models and their Hugging Face checkpoint identifiers.

Model	Checkpoint
Qwen2.5-3B-Instruct	Qwen/Qwen2.5-3B-Instruct
Llama-3.2-3B-Instruct	meta-llama/Llama-3.2-3B-Instruct
Gemma-3-1B-IT	google/gemma-3-1b-it

Each model answered 30 original questions and 90 rewritten versions, producing 360 responses in total. Sampling was disabled (`do_sample=False`), and responses were limited to 32 new tokens (`max_new_tokens=32`). The model-loading configuration used `trust_remote_code=False`.

The generated responses were saved in `outputs/generations/`. A parser then extracted each final answer and compared it with the reference answer. The extracted results were saved in `outputs/parsed/`.

Practical example. For the picture-frame question in Table 2, the correct answer is option E (“table”). Qwen generated E, while Llama generated E. for the original question. The parser extracted E from both responses, so both were counted as correct. The evaluation checks the answer, not whether the response text is written in exactly the same way.

If the parser could not extract an answer, the response was counted as incorrect rather than removed. Qwen had 120 successfully parsed responses, Llama had 118, and Gemma had 119. All three models retained 120 inputs in their accuracy calculations.

4.2 Evaluation Metrics

Three measures were used to examine different aspects of correctness. Let $c_{m,i}^{(t)}$ equal 1 if model m answers question group i correctly under formulation t , and 0 otherwise. The formulations are Original (O), Lexical (L), Syntactic (S), and Information Order (I), with $N = 30$ groups.

Accuracy. Accuracy is the proportion of correct answers in a formulation:

$$A_m^{(t)} = \frac{1}{N} \sum_{i=1}^N c_{m,i}^{(t)}. \quad (1)$$

For example, 15 correct answers out of 30 give 50% accuracy. Changes from original accuracy are reported in percentage points.

Correctness flip rate. A flip occurs when an answer changes from correct to wrong ($C \rightarrow W$) or from wrong to correct ($W \rightarrow C$):

$$F_m^{(t)} = \frac{N_{C \rightarrow W} + N_{W \rightarrow C}}{N}. \quad (2)$$

Both directions are recorded because accuracy can remain unchanged even when different individual questions are answered correctly.

Strict robust accuracy. A question group is counted as robust only if all four versions are answered correctly:

$$R_m = \frac{1}{N} \sum_{i=1}^N \prod_{t \in \{O, L, S, I\}} c_{m,i}^{(t)}. \quad (3)$$

For example, if the original and two rewrites are correct but the third rewrite is wrong, the group does not count as robust. This metric uses 30 question groups, not 120 individual inputs.

4.3 Statistical Analysis

Because the original and rewritten versions belong to the same question groups, paired statistical methods were used.

Two-sided exact McNemar tests compare the numbers of $C \rightarrow W$ and $W \rightarrow C$ changes [McNemar, 1947]. Bootstrap resampling of complete question groups provides 95% confidence intervals for accuracy differences [Efron, 1979].

The Benjamini–Hochberg procedure accounts for multiple testing [Benjamini and Hochberg, 1995]. It was applied across 36 comparisons: three models, three transformation types, and four evaluation scopes (the combined dataset and each individual dataset).

Adjusted p-values were assessed at $\alpha = 0.05$. Results that did not pass this threshold are reported as observed differences in the sample, not as established general effects.

5 Results and Analysis

The three models were evaluated on the same 30 question groups, with four formulations per group. The results below show how accuracy changed, whether individual answers changed correctness, and how often all four versions were answered correctly.

5.1 Accuracy Across Formulations

Across all 120 inputs per model, Qwen answered 60 correctly (50.0%), Llama answered 50 (41.7%), and Gemma answered 27 (22.5%).

Table 4 shows the accuracy of each model for the original and rewritten questions.

Table 4: Accuracy by model and formulation. Each formulation contains 30 inputs; overall accuracy uses 120 inputs.

Model	Orig.	Lex.	Syn.	Info.	Overall
Qwen	63.3	50.0	50.0	36.7	50.0
Llama	50.0	40.0	40.0	36.7	41.7
Gemma	23.3	23.3	20.0	23.3	22.5

All values are percentages.

Information Order produced the largest observed accuracy decrease for Qwen and Llama. Qwen decreased from 63.3% on original questions to 36.7% under Information Order, a difference of 26.7 percentage points. Llama decreased from 50.0% to 36.7%.

Gemma’s accuracy changed little across formulations. However, unchanged accuracy does not

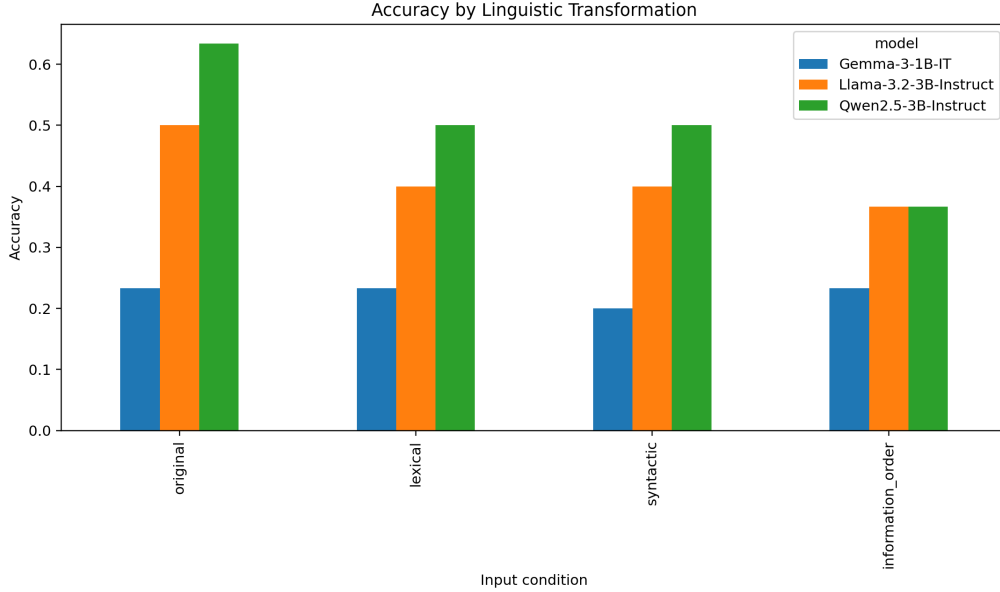


Figure 1: Accuracy on original, lexical, syntactic, and information-order questions.

necessarily mean that the same individual questions were answered correctly.

5.2 Original and Strict Robust Accuracy

Original accuracy counts correct answers to the 30 original questions. Strict robust accuracy counts only question groups in which all four formulations were answered correctly.

Table 5: Original accuracy and strict robust accuracy across 30 question groups per model.

Model	Original	Robust	Gap (pp)
Qwen	19/30 (63.3%)	9/30 (30.0%)	−33.3
Llama	15/30 (50.0%)	8/30 (26.7%)	−23.3
Gemma	7/30 (23.3%)	6/30 (20.0%)	−3.3

For example, Qwen answered 19 original questions correctly, but only nine groups remained correct across all four formulations.

Gemma had a smaller robustness gap, but its original accuracy was also much lower. A smaller gap alone does not indicate stronger overall performance.

5.3 Correctness Flips and a Real Example

A correctness flip occurs when an originally correct answer becomes wrong ($C \rightarrow W$), or an originally wrong answer becomes correct ($W \rightarrow C$).

Table 6 shows both directions for each model and transformation.

Qwen had a 33.3% flip rate under Information Order: nine answers changed from correct to wrong, while one changed from wrong to correct.

Accuracy and flip rate describe different things. Qwen had 50.0% accuracy under both Lexical and Syntactic reformulation, but their flip rates were 13.3% and 26.7%. The number of correct answers was the same, while the individual questions answered correctly were not.

Example from the experiment.

For the picture-frame question in Table 2, the correct answer is E (“table”). Llama produced the following extracted answers:

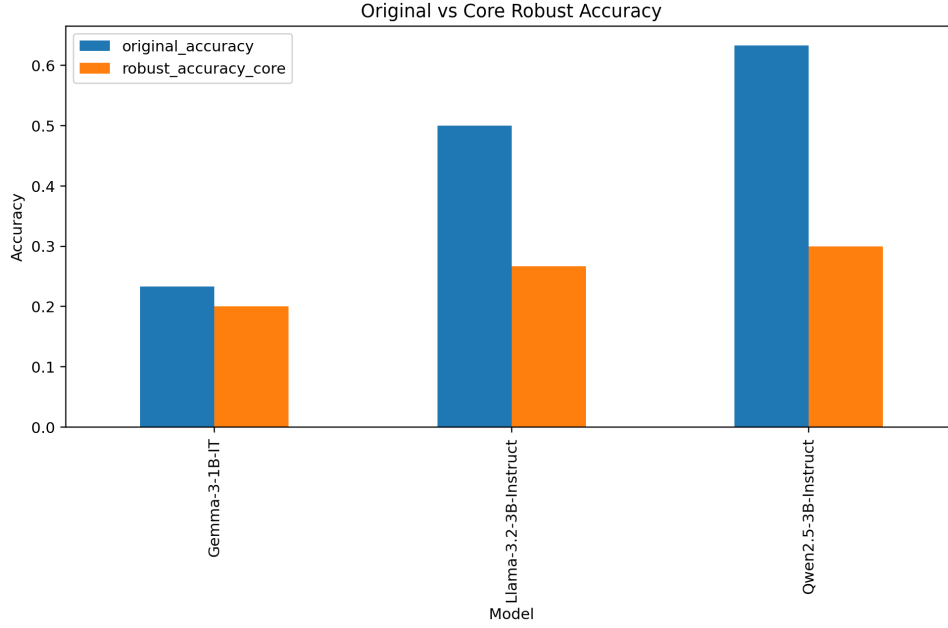


Figure 2: Original-question accuracy compared with the proportion of question groups answered correctly in all four forms.

Table 6: Correct-to-Wrong (C-W) and Wrong-to-Correct (W-C) transitions. Each row compares 30 original and rewritten question pairs.

Model	Transform.	C-W	W-C	Flip (%)
Qwen	Lexical	4	0	13.3
	Syntactic	6	2	26.7
	Info. Order	9	1	33.3
Llama	Lexical	4	1	16.7
	Syntactic	3	0	10.0
	Info. Order	5	1	20.0
Gemma	Lexical	1	1	6.7
	Syntactic	1	0	3.3
	Info. Order	1	1	6.7

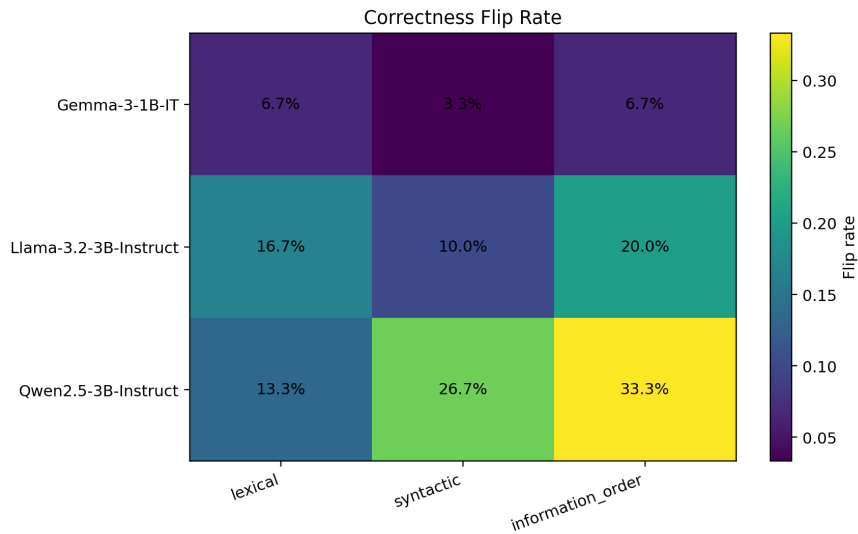


Figure 3: Percentage of question pairs whose correctness changed after rewriting, in either direction.

Table 7: Llama’s answers to the four versions of one real CommonsenseQA question.

Version	Answer	Correct answer	Correct?
Original	E	E	Yes
Lexical	E	E	Yes
Syntactic	E	E	Yes
Information Order	B	E	No

The original, Lexical, and Syntactic answers were correct, but the Information Order answer was wrong. This is one $C \rightarrow W$ flip in the Information Order comparison.

The group also fails strict robust accuracy because all four answers must be correct. This example shows how a model can answer the original question correctly without remaining correct across every rewritten version.

5.4 Statistical Findings

Exact McNemar tests compared the paired correctness changes, while bootstrap confidence intervals estimated uncertainty in accuracy differences. The Benjamini–Hochberg correction was applied across 36 comparisons, as explained in Section 4.3.

For Qwen under Information Order, the exact McNemar test gave $p = 0.021484$. After correction, the adjusted value was $p_{BH} = 0.773438$.

No comparison remained statistically significant after correction at $\alpha = 0.05$. The observed differences therefore describe this small evaluation sample and should not be treated as established effects across other questions or models.

The full test results are available in `results/paired_robustness_results.csv`.

5.5 Findings by Research Question

RQ1 – Performance Change. Accuracy differed across formulations. Information Order had the largest observed decrease for Qwen and Llama in this sample.

RQ2 – Correctness Stability. Both Correct-to-Wrong and Wrong-to-Correct changes occurred. Some conditions had equal accuracy but different individual answers were correct.

RQ3 – Robust Accuracy. Strict robust accuracy was lower than original accuracy for all three models. A correct original answer did not guarantee correct answers to all three rewrites.

6 Discussion and Limitations

6.1 Interpretation of the Findings

The results show that a correct answer to an original question does not guarantee a correct answer when the same question is rewritten. This is consistent with earlier research on linguistic variation and model behavior [Ribeiro et al., 2018; Srikanth et al., 2024].

The findings also show why accuracy alone is not enough. Two formulations can have equal accuracy even when different individual questions are answered correctly. For example, Gemma had equal accuracy under Original, Lexical, and Information Order, but the paired analysis still found correctness flips.

Strict robust accuracy adds another perspective by checking whether a model answers all four versions of a question correctly. Its lower values show that correctness on the original question was not always maintained across the three rewrites.

Information Order produced the largest observed accuracy decrease for Qwen and Llama. However, no comparison remained statistically significant after multiple-testing correction. The results

therefore describe this sample rather than establish that one transformation is generally more difficult.

6.2 Limitations and Future Work

The findings should be considered alongside several limitations.

Sample and model coverage. The experiment uses 30 original question groups, ten from each dataset, and three relatively small open-weight models. The results may not represent the full benchmarks or other model families.

Meaning preservation. All 90 rewrites passed automated structural checks, but only 45 received documented manual semantic review. Although all reviewed rewrites were accepted, the remaining rewrites were not independently checked for full meaning preservation.

Evaluation scope. Only three transformation types were tested. The study measures final-answer correctness rather than internal reasoning or explanation faithfulness. A correctness flip shows that an answer changed, but does not establish why the model produced it.

Statistical uncertainty. The small sample limits the precision of the results. The lack of significant findings after correction does not prove that linguistic changes have no effect.

Future work could use more question groups and models, test additional rewriting types, and manually review every rewrite. A larger and more carefully validated evaluation would help determine whether the observed patterns remain consistent across different questions and models.

7 Conclusion and Reproducibility

7.1 Conclusion

This study examined whether language models maintain correct answers when reasoning questions are rewritten without intending to change their meaning. Three models were tested using original, lexical, syntactic, and information-order formulations.

The results showed both Correct-to-Wrong and Wrong-to-Correct changes. Original accuracy alone could not reveal these changes or show whether correctness was maintained across all four versions. Accuracy, correctness flips, and strict robust accuracy therefore provide complementary views of model performance.

The findings are limited to this small sample. No comparison remained significant after multiple-testing correction, so the observed differences should not be treated as general effects.

7.2 Reproducibility and Research Materials

The dataset, validation records, model settings, generated answers, and results are available in the GitHub repository:

GitHub: <https://github.com/Yash-Gavade/ReasonShift>

The main materials are stored in:

- **Dataset and validation:** data/validated/, annotations/, audit/.
- **Prompts and settings:** prompts/, configs/.
- **Model answers:** outputs/generations/, outputs/parsed/.
- **Results and figures:** results/, figures/.

The saved results can be regenerated using:

```
python -m src.analyze_results
python -m src.generate_figures
```

The README explains the setup, model inference, answer parsing, and reproduction steps.

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Declaration of Academic Integrity

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Thesis title: Same Meaning, Different Answer?

Programme: M.Sc. Natural Language Processing

Examiner: Prof. Dr. phil. Simon Münker

This is an achievement for a

☐ Bachelor's thesis

☐ Master's thesis

☒ other examination achievement for module (term paper etc.) Natural Language Processing (MA2NLP2002)

Trier, 30/09/2026

Place, Date

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